
DADS 6002 / CI 7301

Big Data Analytics

Hadoop Lab

Quickstart Virtual Machine Installation

1. Install Oracle VirtualBox (Virtual Machine) on Window
(Requires RAM at least 8 GB)
 - 1.1 Download VirtualBox from
<https://www.virtualbox.org/wiki/Downloads>
 - 1.2 Install VirtualBox by executing the downloaded file.
2. Download the zip file of the Cloudera Quickstart Machine Image from the MS Teams. Then unzip the file (using 7-zip).
3. Start the VirtualBox and click File->Import Appliance then choose the unzip file (.ovf) from 2. to import the cloudera virtual machine.
4. When the Quickstart completely starts, open a terminal window (click on a terminal icon) to work with the machine.

Share A Folder

- To share a folder of Cloudera virtual machine with Window host machine, do the follows:
 1. Click Devices menu of the above Virtualbox menu bar, then select Shared Folders Setting
 2. Click on Machine Folders and select add folder icon (one with + sign on the right)
 3. Enter the pre-created folder path of the Window host to be shared e.g. d:/vbshare. The folder name can be used as the shared folder name e.g. vbshare. Then select Auto-mount and Make Permanent options.

Share A Foloder

4. On a terminal window of the virtual machine, create a shared directory, e.g.
`# mkdir /home/cloudera/vbshare`
5. Login as Super user or root
`# sudo su -`
6. Mount the shared Window folder using shared folder vbshare to the pre-created directory /home/cloudera/vbshare
`# mount -t vboxsf vbshare /home/cloudera/vbshare`
`# exit`

Hadoop Lab 1

1. Under Quickstart
2. `# cd /home/cloudera`
3. Edit a file, "test.txt", using nano command as follows

`# nano`

Enter multiple words and multiple lines, then press `ctrl o` to save to the file named test.txt, and then `ctrl x` to exit from the nano command. The file will be used for word count problem later.

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5. Execute the following hadoop file system commands.

```
# hadoop fs -help | more
# hadoop fs -put test.txt /user/cloudera/test.txt
# hadoop fs -ls /user/cloudera
# hadoop fs -cat /user/cloudera/test.txt
# hadoop fs -get /user/cloudera/test.txt test1.txt
# hadoop fs -mkdir /user/cloudera/temp
# hadoop fs -cp /user/cloudera/test.txt /user/cloudera/temp/test.txt
# hadoop fs -rm /user/cloudera/temp/test.txt
# hadoop fs -rmdir /user/cloudera/temp
# hadoop fs -ls
```

Hadoop Lab 1

Submit a MapReduce job for Word Count problem

- Download two files, mapper.py and reducer.py, from MS Teams into the shared folder.
- Submit a MapReduce job through Hadoop Streaming

```
# hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar
-input /user/cloudera/test.txt -output /user/cloudera/wc -
mapper "python mapper.py" -reducer "python reducer.py" -
file mapper.py -file reducer.py
```
- Output will be written in multiple files under /user/cloudera/wc e.g. part-00000,

mapper.py :

```
#!/usr/bin/env python
```

```
from operator import itemgetter
import sys
```

```
for line in sys.stdin:
    for word in line.split():
        print(word+"\t1")
```

reducer.py :

```
#!/usr/bin/env python
```

```
from operator import itemgetter
import sys
```

```
curkey = None
```

```
total = 0
```

```
for line in sys.stdin:
```

```
    key, val = line.split("\t", 1)
```

```
    val = int(val)
```

```
    if key == curkey:
```

```
        total += val
```

```
    else:
```

```
        if curkey is not None:
```

```
            print(curkey+"\t"+str(total))
```

```
        curkey = key
```

```
        total = val
```

```
print(curkey+"\t"+str(total))
```